

To create a DNA barcode, a specific region of DNA is sequenced so that it can be compared to an online database of reference DNA. One region of DNA that is commonly used is 648 base pairs long within a mitochondrial gene, coding for the enzyme cytochrome c oxidase I.

DNA barcoding has shown that approximately 2.5 million years ago *T. solitaria* evolved into two subspecies, *Tringa solitaria solitaria* and *Tringa solitaria cinnamomea*.

T. s. solitaria breed in eastern North America whereas *T. s. cinnamomea* breed in western North America.

[4]

(b) Fig. 3.1 shows an American oystercatcher, *Haematopus palliatus*.



Fig. 3.1

The black oystercatcher, *Haematopus bachmani*, has all black feathers.

DNA barcoding analysis suggests that the American oystercatcher and the black oystercatcher are **not** separate species.

Suggest how DNA barcoding evidence could indicate that the American oystercatcher and black oystercatcher are **not** separate species.

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..... [2]

(c) Customs officers at airports can use a hand-held DNA barcoding device to identify biological specimens entering or leaving a country.

Suggest how this helps protect endangered species.

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[Total: 8]